Optimization

pKYC Transformation

Dynamic Monitoring Model

pKYC Proof-of-Concept | 10,000 Customers | 50,000 Transactions

Prepared by: Chan Chin To
Date: 4 May 2026

Disclaimer:

All datasets used in this project are 100% synthetic dummy data generated exclusively for Proof of Concept (PoC) and demonstration purposes. No actual customer data or personally identifiable information (PII) is involved, in strict compliance with data privacy regulations.

TOO LONG, DIDN’T READ.

Perpetual KYC (pKYC) is a **continuous, technology-driven approach to customer due diligence** that replaces periodic reviews with **real-time monitoring and automated updates of customer data**. It is designed to improve AML compliance, reduce fraud risk, and increase operational efficiency by enabling financial institutions to detect changes in customer risk profiles as they happen.

With the right solution, firms can streamline compliance processes, reduce manual workloads, and support faster, more **informed decision-making throughout the client lifecycle.**

CHALLENGE: INEFFECTIVE PERIODIC REVIEW

Redefining Compliance from "Time-driven" to "Data-driven" intelligence.
Analysis based on Demo Data with 10,000 Customers and 50,000 Transaction in 30 Days record.

Data Evidence

Ineffective workload

10,000 cases involve Ghost AC/ZBA accounts, yet they still require periodic review.

Time lag

Review are conducted every 1/2/3 years, making it impossible to detect risk changes in real time during the intervals between reviews.

Repeat

A large volume of low-risk cases involve merely copying and pasting the data repeatedly.

Estimation

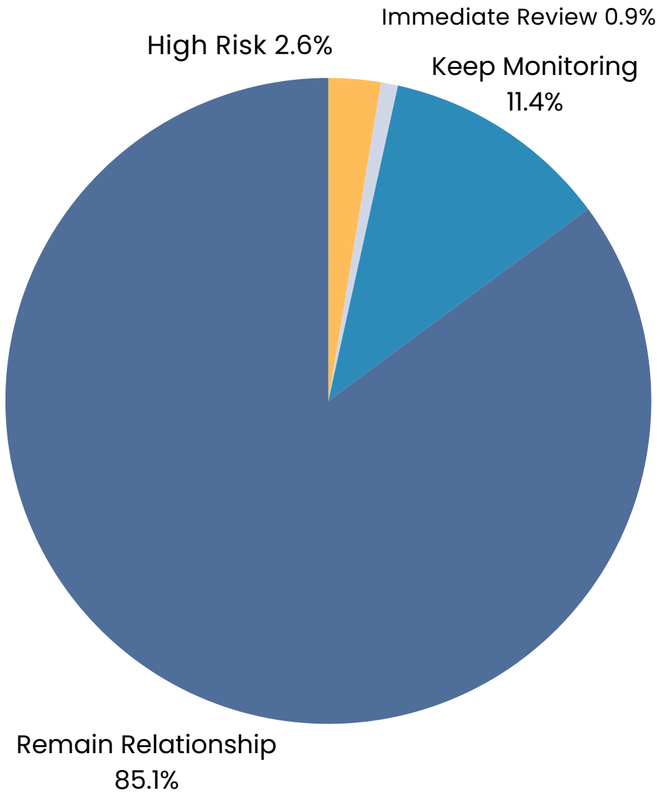
84
Immediate Review

69%
No Action

90%
Save Working hours

SCALE OF RESOURCE MISALLOCATION

Analysis based on 10,605 Review cases from MIS report, including Periodic review and Trigger event.



THE PARADIGM SHIFT

FROM STATIC TO PERPETUAL

FROM STATIC TO PERPETUAL

Traditional Periodic Review

- Time-driven (1/2/3 years)
- High manual effort on low-risk clients
- Retrospective & Reactive
- **Outcome: Inefficient Backlogs**

Perpetual KYC (pKYC)

- Event-driven (Trigger-based)
- Zero-touch for Ghost/Low-risk segments
- Real-time & Proactive
- **Outcome: Resource Optimization**

THE "BRAIN" OF pKYC: BEHAVIORAL TRIGGERS

Moving beyond static documents to monitor living client personas.

Transactional

Trigger review
If Txn patterns deviate from historical baseline

Interactional

App behavioral and Campaign interests
Analysis the client's App logins history, CS Call, and Activity in Campaign.

Product-based

1) Distribution of Client Assets

2) Income Group

3) Occupation

Strategic Outcome: 90% of "Good" customers experience zero-friction compliance, while resources focus on the 3.5% of actual High-Risk threats.

CORE VALUE: DYNAMIC CDD

Differentiated Automated Strategies by Client Segment)

High Risk

Continuously monitor AUM and abnormal transaction frequency. If a major event is triggered, initiate review within 24 hours.

Medium and Low Risk

Change periodic reviews. Intervene only when triggered by events such as job change, adverse news, or cross-border address changes.

Zero-Value

Fully exempt from the review process. System automatically flags them for batch closure or long-term dormancy. This saves 29% of resources.

Real-Time Data Monitoring

Risk Filtering

Automated Completion (Low Risk)

Focus on High Risk

5-DIMENSION ALERT FRAMEWORK

Identifying administrative redundancies to focus on high-value risk management.

1

Risk Score Alert Triage

Combines country risk + external trigger score with initial risk rating

Score ≥ 70 → High Risk | Score = 0 & High Initial → Over Rating

2

Transaction Pattern vs Income

Compares total transaction amount against stated monthly income

Ratio > 10x AND Count > 10 → High Risk

3

Threshold Evasion (Structuring)

Identifies transactions just below the $10,000 reporting threshold

≥5 near-threshold txns → High Risk | 2–4 → Warning

4

Geographic Anomaly

Low-income (Entry) customers with high-risk country exposure

Country Score ≥ 90 → Exit | 70–89 → High Risk | 30–69 → Warning

5

Alert Synergy (Final Priority)

Consolidates all 4 alert dimensions into a single action priority

Score ≥ 5 → Immediate Review | 3–4 → High Risk Review | 2 → Monitor

MODEL RESULTS

(10,000 Customers)

⚠ Highest Priority: Geographic risk exposure signals an urgent need for enhanced due diligence on high-risk jurisdiction transactions.

Action 1 — Risk Triage

332 'Over Rating' (conservative)

11.9 %

Action 2 — Txn Pattern

90.4% normal transaction pattern

0.4 %

Action 3 — Structuring

No High-Risk cases detected

1.3 %

Action 4 — Geographic

35 customers require immediate exit

7.3 %

Action 5 — Final

3.5 %

CASE RESULT (DEMO)

Customer_ID	Description	Risk Score	Final Action
C_08371	Low income + Geographic anomaly (Exit) + Abnormal transaction volume (High)	**5**	**Immediate Review**
C_02327	High-rating alert + Abnormal transaction volume (High)	**4**	High Risk Review
C_00001	Single-dimensional high-risk alert	2	Keep Monitoring

VISION

We should no longer waste resources on **repetitive Copy and Paste tasks for the 90% of 'Good' Customers.** Instead, through perpetual KYC (pKYC), we can free up manpower and transform the compliance team into **true data-driven risk analysts.**

NEXT STEP

Efficiency Improvement

New Reviews path
For the low-risk client group review costs are reduced by up to 90%.

Backlog Reduction
Only truly necessary cases are processed, minimizing accumulation through Dynamic CDD.

Risk Control Quality Enhancement

Timeliness
Shift from "post-event review" to "prevent / in-process monitoring", turning compliance proactive.

*Deloitte 2025 studies report 60% improvement in early risk detection

Precision
Resources concentrated on 3.5% high-risk individuals, boosting AML detection rates.
Enhances quality, reduces oversight errors, and strengthens regulatory defensibility.

ROAD MAP

NOW

Deploy Python model to GCP

Implement Isolation Forest algorithm

Full automation

MOVING TOWARD INTELLIGENT MONITORING

FROM STATIC TO PERPETUAL

CUSTOMER EXPERIENCE

REACH PATH CHANGE

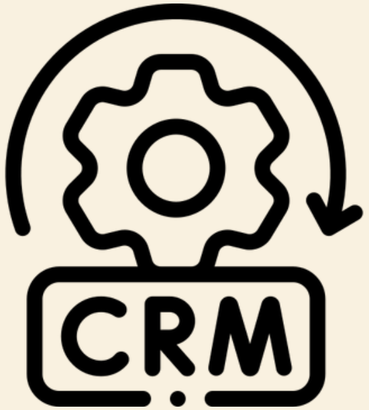
Allocating reaching resources based on customer value and risk tier.

Annoying

- **A pop-up appears immediately** upon logging into the app, requiring customers to answer questions and disrupting their ability to use the app.
- **Emails** sent to customers requesting information come across as **highly unprofessional**.
- Banks **repeatedly call and send SMS/emails** to urge customers: “Please update your information as soon as possible, otherwise your account may be restricted.”

Customers Experience improvement

switching to Push Notifications or an in-app message inbox to contact customers for KYC updates.
it significantly reduces customer friction, complaints, and attrition.



CRM System
Centralized Customer Profile

SYNERGIZING CDD WITH CRM

Unlocking the "Single Source of Truth" to enrich customer profiles

Data Enrichment

Real-time Sync:
Feed occupation, income, and life-stage data from CDD updates directly into CRM Profiles.

Accuracy:
CDD is the only touchpoint where data is legally verified, ensuring CRM remains the **Golden Record.**

Preference Tracking:
Capture digital behavior (App usage, response time) to **refine Customer segmentation**.

Integrated Workflow

360° View:
Sale Team gain full visibility into a client's CDD status within the CRM interface.

Zero Redundancy:
Stop asking clients for information the bank already holds in CRM.

Cost Reduction:
100% data alignment prevents "Failed Rejections" caused by data discrepancies between systems.

TURNING COMPLIANCE INTO GROWTH

Converting mandatory CDD touchpoints into intelligent Sales Leads.
Deliver a better customer experience, reduce friction , enhance satisfaction, increase customer retention.

High-Intent Interaction

Client logs in to update CDD info
Use the final "Thank You" screen and Cross-sell customized TD offers or Insurance products.

Career Advancement Trigger

System detects occupation change during **dCDD**
Auto-trigger CRM Lead for Credit Card / Loan / TU report

Life-Stage Trigger

Change of address or marital status captured
Offer Mortgage or Wealth Management Health Checks.

CRM CORE FUNCTIONS

Key system capabilities required to support the pKYC-CRM bridge.

Lead Scoring

Sales teams focus only on high-scoring leads (top 20%), reducing wasted time and resources.

Based on the **CDD Risk Score** and **AUM Potential**. Focus Sales efforts on high-yield profiles.

Next Best Action

AI analyzes customers' latest behavior and data in real time, automatically recommending the "best next action" — either pushed to the RM or delivered directly to the customer via the app/PN.

Closed-Loop Feedback

Marketing and Sales DATA lead into CRM. Results are fed back to marketing to automatically optimize scoring, NBA (Next Best Action), and campaigns.

Marketing no longer has to guess which channel is effective; instead, it refines strategies based on actual sales results → higher ROI and more precise targeting.

Identify high-value customers + **Instantly recommend the right product** + **Continuously learn and optimize**

Requirement: Seamless API integration between the pKYC Engine and CRM to ensure zero latency in data synchronization.
Result: Compliance becomes a revenue engine, not just a cost center.

CRM RECOMMENDATION

Salesforce Financial Services Cloud

Microsoft Dynamics 365

HubSpot / Zoho CRM

RISK ASSESSMENT

Proactive identification of potential hurdles and strategic mitigation plans.

Risk	Impact	Challenge	Mitigation
Legacy Tech	HIGH	CBS lack real-time APIs to feed the pKYC Engine.	**Trigger Monitoring**: NOT stopping monitoring. We are switching to "Event-Driven".
HKMA	HIGH	"Removing Periodic Review for Ghost Accounts create a risk vacuum for money laundering?	**Parallel Run Period:** Run pKYC and traditional PR for 6 months to prove to Audit that no risks were missed before full switch-over.
Data Privacy	MED	Feeding sensitive CDD data into CRM exposes privacy risks.	**View Only:** CRM only displays actionable flags, keeping sensitive PII documents locked in the CDD core system.

IMPLEMENTATION ROADMAP

Phased Transition Strategy: From Cleanup to Value Generation

1 Data Purge

TARGET

10,000 Customers
50,000 Txn Records

Action

Demo analysis the Data to test the logic and program.

Outcome

Immediate 90% resources reduction.

2 Pilot Test

TARGET

Project Study

Action

Test 'New Reach Path' (PN / In-APP msg) and Keep Monitoring this batch customers.

Outcome

Validate STP Logic & Clear 3.5% High Risk Case.

3 Tech & CRM

TARGET

IT, Business, FCC, Risk

Action

Deploy API middleware to feed CDD triggers into CRM.

Outcome

Enable 'Lead Scoring' & 'Next Best Action'.

4 pKYC Rollout

TARGET

Mass Market (Total Cases)

Action

Decommission Periodic Review. Go live with Data-Driven Monitoring.

Outcome

365-Day Real-time Risk Visibility.

Q & A

Appendix 1 – Information

Row Labels	Count of Customer_ID
High	996
Low	6926
Medium	2078
Grand Total	**10000**

Row Labels	Count of Customer_ID
High	**996**
High Risk Review	154
Immediate Review	10
Keep Monitoring	500
Remain Relationship	332
Low	**6926**
High Risk Review	82
Immediate Review	56
Keep Monitoring	492
Remain Relationship	6296
Medium	**2078**
High Risk Review	25
Immediate Review	19
Keep Monitoring	149
Remain Relationship	1885
Grand Total	**10000**

Income Group	Low	Medium	High	Grand Total
Elite(>100k)	**2468**	**749**	**351**	**3568**
High Risk Review	14	2	10	26
Keep Monitoring	129	40	218	387
Remain Relationship	2325	707	123	3155
Entry(<=50k)	**1858**	**541**	**281**	**2680**
High Risk Review	52	13	128	193
Immediate Review	56	19	10	85
Keep Monitoring	220	55	41	316
Remain Relationship	1530	454	102	2086
Middle(50k-100k)	**2600**	**788**	**364**	**3752**
High Risk Review	16	10	16	42
Keep Monitoring	143	54	241	438
Remain Relationship	2441	724	107	3272
Grand Total	**6926**	**2078**	**996**	**10000**

Appendix 2 – Action

Triage Action	Customer Count	% of Total
Normal	8,482	84.8%
High Risk	1,186	11.9%
Over Rating	332	3.3%

	Geog_Anomaly_Analysis	Customer_Count
0	Warning	929
1	Normal	626
2	High Risk	88
3	Extremely High Risk / Exit	35

Pattern Category	Customer Count	% of Total
Normal	9,036	90.4%
Warning	924	9.2%
High Risk	40	0.4%

Final Priority	Customer Count	% of Total
Immediate Review	84	0.8%
High Risk Review	335	3.4%
Keep Monitoring	2,682	26.8%
Remain Relationship	6,899	69.0%

Structuring Analysis	Customer Count	% of Total
Normal	9,871	98.7%
Warning	129	1.3%

Reference

•Deloitte (2025). Perpetual KYC Models: Insights on Early Risk Detection. Banks adopting pKYCachieved 60% improvement in early risk detection.
Source: https://www.partisia.com/blog/perpetual-kyc-pkyc-transforming-compliance-from-periodic-to-continuous

•PwC (2024-2025). Perpetual KYC: A New Approach to Periodic Reviews. pKYC reduces KYC maintenance costs by up to 40%; medium-sized banks save ~USD 14.4M annually.
Source: https://www.pwc.com/sg/en/consulting/assets/pkyc-a-new-approach-to-periodic-reviews.pdf

•HKMA (2024-2025). Guidance on Risk-based Approach for Customer Due Diligence & Ongoing AML/CFT Controls. Supports event-driven perpetual monitoring over rigid periodic reviews.
Source: https://www.hkma.gov.hk/eng/key-functions/banking/anti-money-laundering-and-counter-financing-of-terrorism/guidance-papers-circulars/

•Fenergo (2025). Perpetual KYC Guide to Ongoing Due Diligence. Real-time behavioral monitoring for efficiency and risk management.
Source: https://resources.fenergo.com/blogs/perpetual-kyc-pkyc